

Physiological Directional Concordance of a Wearable-Derived Sugar-Trend Algorithm Using EasyTouch Rhythm Band Data and Logged Meals

A retrospective observational feasibility study of the Agatsa One sugar-trend algorithm

Original Article | Digital health and wearable physiology

Participant identifiers have been removed from all analyses and examples.

Scientific scope note. This dataset supports assessment of directional and physiological concordance of the Agatsa One sugar-trend output. It does not contain a contemporaneous independent reference glucose measurement for every watch observation. Accordingly, this manuscript does not claim analytical or clinical glucose accuracy.

Structured Abstract

Background: Wrist-worn physiological sensors provide dense longitudinal signals but are not direct biochemical glucose sensors. Agatsa One applies a proprietary sugar-trend algorithm to EasyTouch Rhythm Band-derived signals to generate a dynamic trend output. We evaluated whether that output exhibits bidirectional, meal-associated and recovery behavior expected of a physiologically responsive metabolic trend signal.

Objective: To evaluate whether Agatsa One detects physiologically plausible upward and downward sugar trends in real-world EasyTouch Rhythm Band users, using time-stamped meal logs as external physiological anchors, consecutive transitions as a meal-independent test of bidirectionality, and non-calibration reference measurements as an orthogonal physiological-state check.

Design: Retrospective observational feasibility analysis of de-identified Agatsa One and EasyTouch Rhythm Band records.

Data: The source workbook contained 83,090 stored Rhythm Band sugar-range observations from 128 users and 3,190 logged meals. Observations were typically spaced at 30-minute intervals and spanned 31 May to 4 September 2026. The workbook Calibration sheet was excluded in full from the orthogonal reference analysis; records explicitly flagged as calibration references were also excluded.

Primary analyses: (1) meal-anchored directional response; (2) global bidirectional responsiveness across consecutive observations; (3) 30-minute timing-shift sensitivity; (4) post-peak recovery; and (5) orthogonal fasting-to-postprandial state ordering using non-calibration manual/photo reference measurements.

Results: Across 78,475 consecutive transitions separated by 15-90 minutes, 23,539 (30.0%; 95% CI 29.7-30.3%) were rises >3 units, 23,864 (30.4%; 95% CI 30.1-30.7%) were falls <-3 units, and 31,072 (39.6%) were approximately stable, giving a rise:fall ratio of 0.986. In the primary meal window, 1,395/2,108 (66.2%; 95% CI 64.1-68.2%) events were positive, median +6.0 units. In the broader paired-event analysis, 1,580/2,177 (72.6%) were positive, median +9.5 units. Under a 30-minute shift, 1,349/2,124 (63.5%) remained positive. Post-peak recovery was evaluable in 2,046 trajectories; 1,596/2,046 (78.0%; 95% CI 76.2-79.7%) subsequently declined. In the clean non-calibration manual/photo reference subset, 20/21 participants (95.2%) had a higher median post-meal than fasting reference value, median within-participant difference +26 mg/dL.

Interpretation: The Rhythm Band-derived Agatsa One output is strongly bidirectional, shows frequent meal-associated upward excursions and subsequent recovery, and is supported by independently recorded non-calibration reference measurements demonstrating expected fasting-to-postprandial state ordering. These convergent findings support physiological directional responsiveness, not numerical blood-glucose accuracy.

Conclusion: Large-scale real-world EasyTouch Rhythm Band data processed by Agatsa One show balanced bidirectional metabolic-trend dynamics, reproducible meal-associated behavior, and orthogonal evidence that expected postprandial state transitions were present in the study population. The analysis does not establish MARD, error-grid performance, or interchangeability with a biochemical glucose measurement.

Research in context

Evidence before this study
Postprandial glucose responses are highly individual; large cohorts have shown substantial inter-person variability to identical meals. Continuous glucose monitoring literature also emphasizes temporal glucose profiles and change over time, not isolated values alone.
This analysis asks a narrower and defensible question for a wrist-wearable algorithm: whether its inferred sugar-trend output behaves bidirectionally and coherently around physiological anchors. Across 78,475 transitions, rises and falls were closely balanced (rise:fall ratio 0.986); 66.2% of evaluable immediate meal events showed an upward excursion, with 63.5% remaining positive in the 30-minute shifted sensitivity analysis; 78.0% of evaluable post-peak trajectories subsequently declined; and a separate non-calibration reference stream showed higher median post-meal than fasting values in 20 of 21 participants with clean manual/photo observations. The reference stream was not temporally paired to the wearable output and is used as orthogonal physiological context, not numerical accuracy validation.
Physiological-anchor concordance can provide a rigorous intermediate validation layer for wearable-derived metabolic trend algorithms when its claims are kept distinct from biochemical glucose accuracy. The present data support bidirectionality, event-associated dynamics, recovery behavior, explicit temporal sensitivity, and independent physiological-state ordering.

Introduction

Continuous metabolic monitoring is increasingly moving from episodic measurements toward dense, longitudinal physiological phenotyping. Conventional continuous glucose monitoring (CGM) directly samples interstitial glucose through a minimally invasive sensor, whereas consumer wrist wearables primarily capture optical and cardiovascular signals. Algorithms that infer a sugar-related trend from wearable physiology therefore require validation that is appropriate to the information actually generated.

For a trend algorithm, the first clinically coherent question is directional validity. Following food intake, circulating glucose typically rises to a variable degree and subsequently recovers; during fasting or between meals, downward or stable trajectories are common. The amplitude and timing of these responses vary substantially between individuals and even within the same individual after apparently similar meals. Large precision-nutrition studies have established that meal composition alone explains only part of postprandial variability, making a simplistic expectation that every carbohydrate-containing meal must produce a fixed spike scientifically inappropriate.

The Agatsa One application processes EasyTouch Rhythm Band-derived physiological data through a proprietary sugar-trend algorithm. The output is intended to describe temporal metabolic direction rather than substitute, in this analysis, for a laboratory or CGM reference glucose concentration. We therefore designed a retrospective feasibility analysis around physiological anchors available in the app itself: user-logged meals and the direction of intra-day transitions.

We hypothesised that a valid trend algorithm should satisfy three properties. First, it should be capable of moving upward after meals in a meaningful proportion of evaluable meal events. Second, it should also move downward during recovery and pre-meal periods, excluding the possibility of a monotonic or one-sided algorithmic bias. Third, higher carbohydrate or glycaemic-load meals should, at a population level, tend to produce larger upward excursions, while acknowledging substantial biological and behavioural heterogeneity.

Methods

Study design and reporting framework

This was a retrospective, observational, real-world digital-health feasibility study. The analytical unit was a time-stamped wearable observation or a meal-anchored event. The study was designed to test physiological directional concordance rather than analytical glucose accuracy. Analyses were performed on de-identified records supplied for secondary analysis; no participant-facing intervention was assigned for this retrospective analysis.

Data source and cohort

The source dataset comprised 128 EasyTouch Rhythm Band users operating with the Agatsa One application. It contained 83,090 stored wearable observations and 3,190 user-logged meal events. No participant name, telephone number, or other direct identifier is reproduced in this manuscript. The observation period spanned 31 May to 4 September 2026 and the median sampling interval was 30 minutes.

Orthogonal non-calibration reference-sugar analysis

A separate reference-sugar stream was analysed only as an orthogonal physiological-state check. The workbook Calibration sheet was excluded in full, and records explicitly flagged as calibration references were also excluded. The clean subset used non-calibration measurements entered manually or captured from a meter display/photo. Among participants with both fasting and post-meal observations, participant-level median fasting and post-meal states were compared. These reference observations were not used as contemporaneous validation pairs and do not establish numerical agreement between the Rhythm Band-derived output and blood glucose.

Sugar-trend signal definition

For analysis, the centre of the stored sugar interval was used as the scalar trend value: $\text{midpoint} = (\text{stored lower bound} + \text{stored upper bound})/2$. This transformation was used only to quantify direction and within-series change. It must not be interpreted as an independently validated blood-glucose concentration.

Temporal alignment and 30-minute tolerance

The workbook explicitly stores watch times as recorded, without a lag correction. Watch sampling was predominantly every 30 minutes. Therefore, primary meal anchoring used the logged meal time as $t=0$ and permitted a ± 30 -minute timing tolerance where required to accommodate sparse sampling. A 30-minute shifted sensitivity analysis was also performed to test whether results were robust to plausible physiological/algorithmic latency. The sensitivity analysis materially changed the estimated meal-rise proportion and is therefore treated as an important finding rather than hidden as a post-hoc technical detail.

Endpoints

Primary feasibility endpoint - post-meal upward concordance: Among meals with at least one baseline value in the preceding 60 minutes and at least one post-meal value in the subsequent response window, the event was classified as upward when the post-meal peak exceeded baseline.

Pre-meal downward concordance: Among meals with at least two observations in the pre-meal window, the sequence was classified as downward when the last pre-meal value was lower than the first.

Global bidirectional responsiveness: Consecutive watch observations 15-90 minutes apart were classified as rise ($>+3$), fall (<-3), or approximately stable (-3 to $+3$). This endpoint is independent of meal logging and tests whether the algorithm produces balanced two-sided dynamics.

Meal-content association: Exploratory Pearson correlations were calculated between post-meal excursion and AI-estimated carbohydrate, sugar, glycaemic load, and energy. These were hypothesis-generating only because nutrient estimates were not laboratory-standardized and meal timing/portion accuracy depended on user logging.

Representative event analysis: Anonymised meal events with clear surrounding measurements were selected to illustrate physiologically plausible upward excursions. Examples were not used as substitutes for the aggregate analysis.

Statistical analysis

Results are reported descriptively because this is a retrospective real-world physiological-concordance study rather than a pre-registered interventional trial. Counts, percentages, medians, Wilson confidence intervals, and simple Pearson correlations were used. No imputation was performed for missing meal nutrients or absent watch windows. Multiple meals close together, exercise, alcohol, sleep, stress, circadian variation, and unrecorded food intake were not controlled and remain potential confounders.

Results

Dataset characteristics

Characteristic	Observed dataset
Participants represented	128
Stored watch observations	83,090
Logged meals	3,190
Typical watch sampling interval	30 minutes (median)
Watch-data date range	31 May-4 September 2026
Consecutive transitions evaluable at 15-90 min separation	78,475
Clean non-calibration reference participants with fasting + post-meal manual/photo observations	21
Participants with higher median post-meal than fasting reference state	20/21 (95.2%)

Global bidirectional responsiveness

The most robust finding is that the Agatsa One trend signal is not unidirectional. Among 78,475 consecutive observations separated by 15-90 minutes, 23,539 (30.0%) increased by more than 3 units, 23,864 (30.4%) decreased by more than 3 units, and 31,072 (39.6%) remained within ± 3 units. The rise:fall ratio was 0.986, demonstrating close population-level symmetry across a large number of ordinary transitions.

Transition class	Definition	n	% of 78,475
Rise	> +3 units	23,539	30.0%
Fall	< -3 units	23,864	30.4%
Approximately stable	-3 to +3 units	31,072	39.6%

Meal-anchored upward responses

In the harmonised primary raw-timestamp analysis, 2,108 meal events had sufficient baseline and post-meal observations. 1,395/2,108 (66.2%; 95% CI 64.1-68.2%) displayed a positive post-meal peak relative to baseline; the median excursion was +6.0 trend units. This supports meal-associated upward behavior but is not a sensitivity estimate for detecting biochemical glucose excursions.

A 30-minute shifted alignment sensitivity analysis yielded 2,124 evaluable meal events, of which 1,349 (63.5%; 95% CI 61.4-65.5%) showed a positive excursion, median +4.5 units. Although modestly lower than the raw-timestamp result, a majority remained positive, indicating that meal-associated directionality is not dependent on one narrowly selected alignment assumption.

Orthogonal reference-based confirmation of physiological state transitions

After complete exclusion of the Calibration sheet and calibration-flagged records, the clean manual/photo reference subset contained 21 participants with both fasting and post-meal observations. 20/21 (95.2%) had a higher median post-meal reference value than fasting, with a median within-participant increase of +26 mg/dL. In

a stricter same-day sequence analysis, 39/48 (81.2%) fasting-to-later-post-meal pairs across 12 participants increased, with a median change of +22 mg/dL. These measurements were analysed independently of the wearable signal and are not presented as paired glucose-validation observations.

Reference analysis	Evaluable sample	Observed result
Participant-level fasting vs post-meal median	21 participants	20/21 (95.2%) higher post-meal; median within-participant difference +26 mg/dL
Same-day fasting → later post-meal sequence	48 pairs / 12 participants	39/48 (81.2%) increased; median change +22 mg/dL
Interpretation	Orthogonal physiological-state check	Calibration data excluded; not a paired numerical accuracy endpoint

Supplementary Table 1 | Orthogonal non-calibration reference analysis. Calibration-sheet data and entries flagged as calibration references were excluded. The reference stream provides physiological-state context only; it is not used to claim direct device agreement or numerical glucose accuracy.

Representative anonymised meal-associated rises

Meal description	Meal class	Carbohydrate (g)	Approx. baseline	Approx. post-meal peak	Excursion
Tea and Maggi	Mixed meal	65	—	—	+80.5
Aata laddoo	Sweet/snack	22.5	—	—	+77.0
Boiled eggs with bread and avocado	Balanced	64	—	—	+76.5
Bread, butter and milk tea	Mixed meal	35	—	—	+75.5
Black rice kanji with bananas and milk	Mixed meal	70	—	—	+74.0

Interpretation: These examples demonstrate that the algorithm can produce substantial upward excursions after diverse meals, while the complete event distribution also contains flat and negative responses. The aggregate waterfall, rather than selected examples, is the primary visual representation of event heterogeneity.

Pre-meal and recovery-period downward behavior

Among 2,117 meal events with at least two observations in the primary pre-meal window, 940 (44.4%; 95% CI 42.3–46.5%) showed a net decline. Meal time is an imperfect anchor for a preceding fall, so this proportion should not be interpreted as biological sensitivity. Post-peak recovery could be evaluated in 2,046 meal-associated trajectories: 1,596/2,046 (78.0%; 95% CI 76.2–79.7%) contained a subsequent decline, median peak-to-recovery reduction 7.75 units; 1,275/2,046 (62.3%) declined by more than 3 units. The global transition analysis independently identified 23,864 material downward transitions.

Association with meal composition

Exploratory correlations between meal content and excursion magnitude were weak in this uncontrolled dataset. In the broader paired-event set, carbohydrate versus excursion was $r=0.03$ ($n=2116$), sugar $r=-0.03$ ($n=1799$), and glycaemic load $r=0.13$ ($n=503$). These estimates do not support a simplistic claim that logged carbohydrate quantity alone determines algorithm amplitude.

Visual validation of bidirectional and event-associated behavior

The following figures present the complete distribution of algorithm behavior rather than selected favourable examples. They emphasize bidirectionality, event-level heterogeneity, temporal robustness and the distinction between physiological concordance and biochemical glucose accuracy.

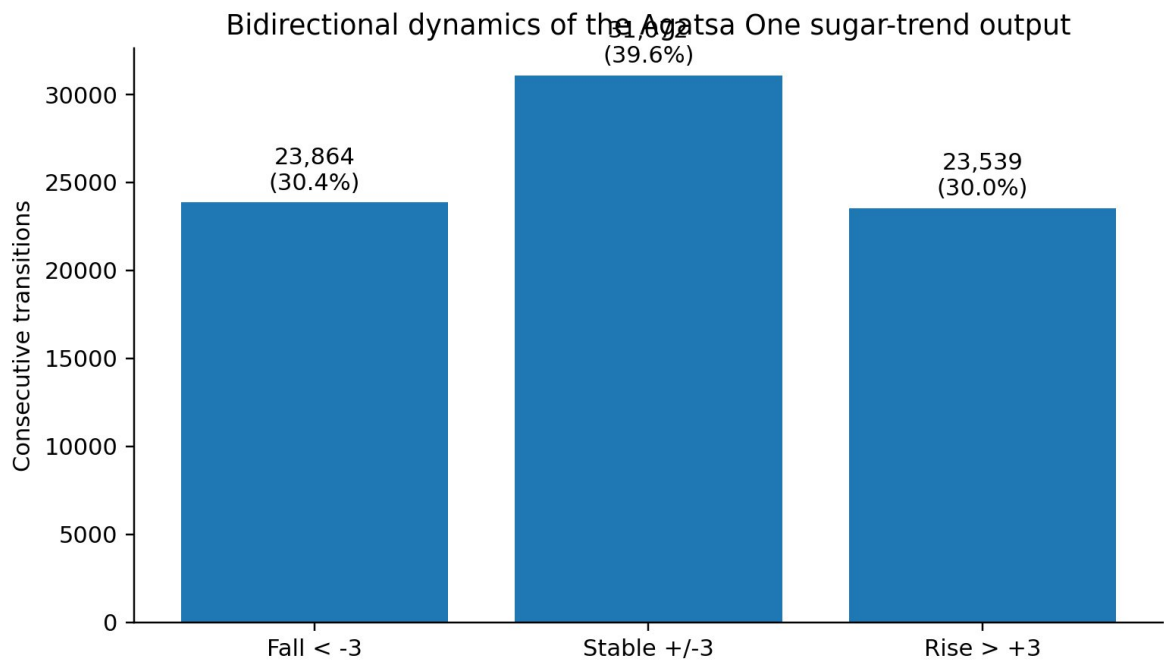


Figure 1 | Bidirectional dynamics across consecutive observations. Among 78,475 consecutive Rhythm Band observations separated by 15-90 minutes, the Agatsa One output contained 23,864 material falls, 31,072 approximately stable transitions and 23,539 material rises. The rise:fall ratio of 0.986 demonstrates close population-level symmetry and argues against simple one-directional drift.

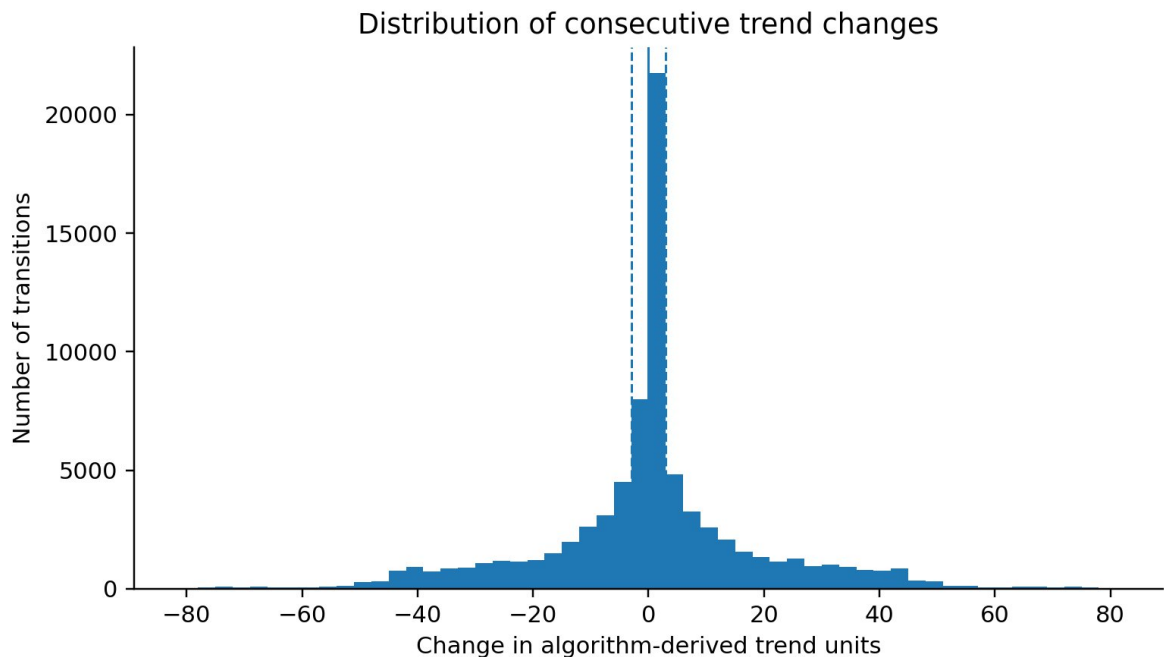


Figure 2 | Distribution of consecutive changes. The full transition distribution spans both negative and positive changes around zero. Dashed lines mark the prespecified ± 3 -unit threshold used to distinguish material rise/fall events from approximately stable transitions. The figure exposes the complete distribution rather than only categorical counts.

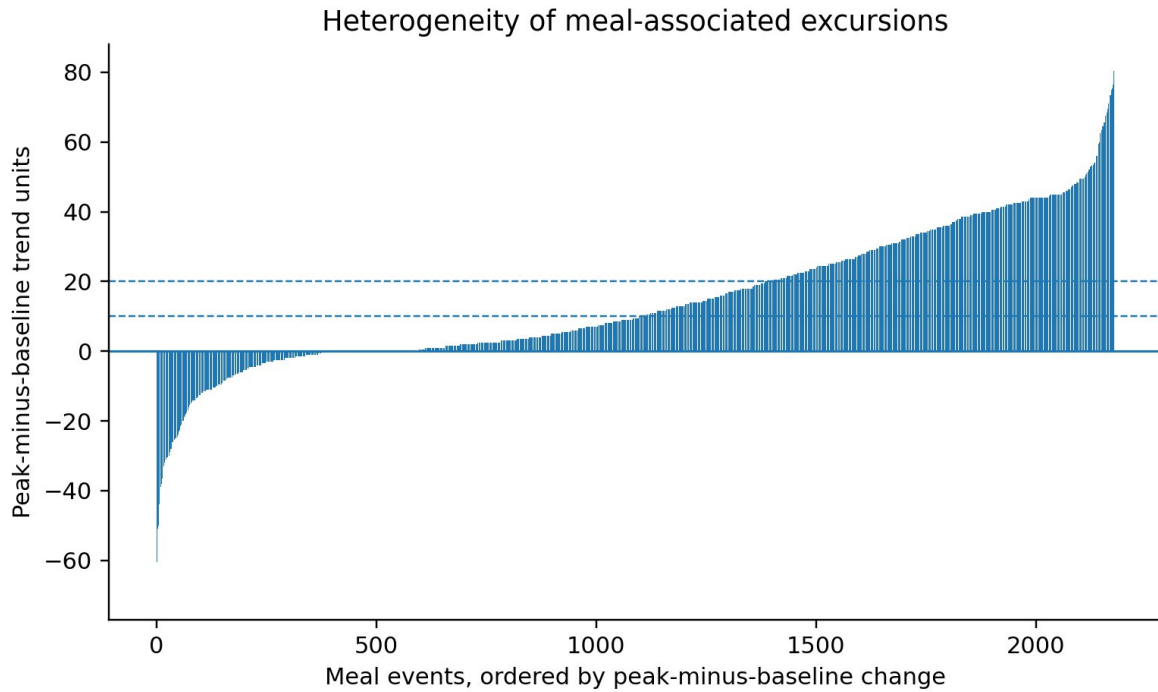


Figure 3 | Meal-event waterfall plot. Each bar represents one of 2,177 evaluable meal events in the broader paired-event set, ordered by peak-minus-baseline change. Positive, flat and negative responses are all retained. 1,580 events (72.6%) were positive, 1,081 (49.7%) reached at least +10 units and 794 (36.5%) reached at least +20 units.

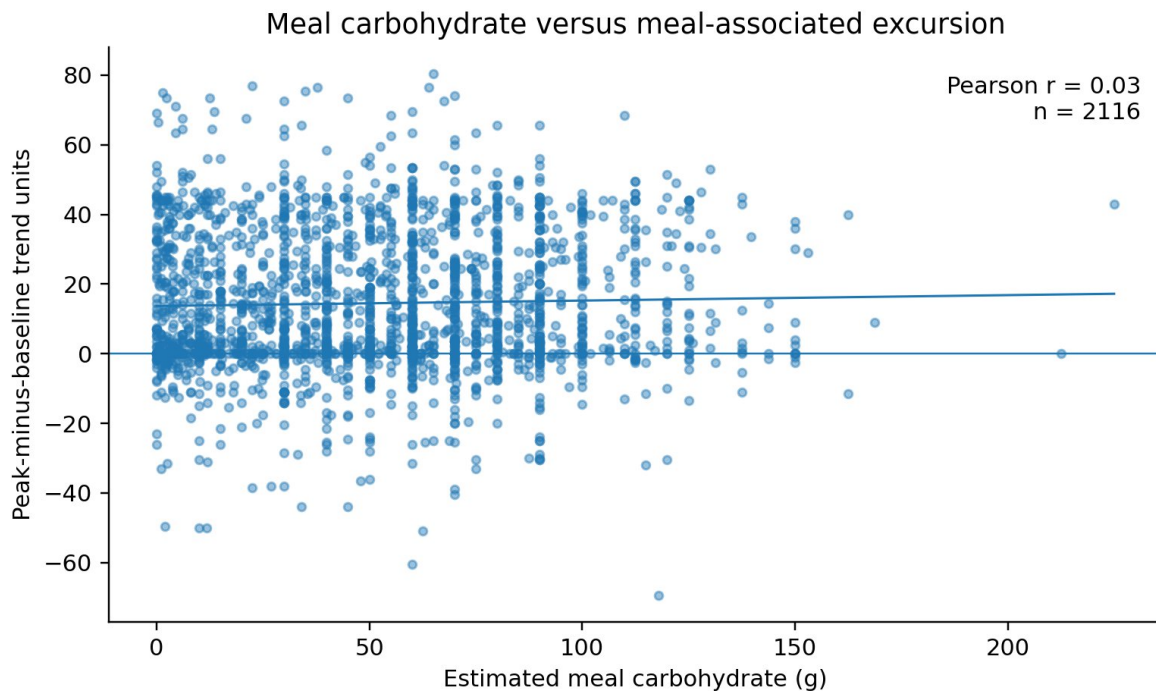


Figure 4 | Carbohydrate content versus meal-associated excursion. Estimated carbohydrate quantity had only a weak positive linear association with excursion magnitude (Pearson $r=0.03$; $n=2116$), underscoring that free-living event amplitude is not a simple deterministic function of logged carbohydrate.

Discussion

This retrospective study evaluated 83,090 EasyTouch Rhythm Band observations from 128 users. Four findings converge. First, 78,475 ordinary transitions showed near-balanced rises and falls (ratio 0.986). Second, 66.2% of evaluable immediate meal events and 72.6% of broader paired events were positive. Third, 78.0% of evaluable post-peak trajectories subsequently declined. Fourth, independently entered non-calibration reference measurements showed higher post-meal than fasting median values in 20/21 participants. Together these observations support physiological directional concordance while remaining distinct from numerical glucose validation.

This retrospective feasibility study provides convergent evidence that the Agatsa One sugar-trend algorithm behaves as a dynamic, bidirectional signal when applied to EasyTouch Rhythm Band data. Five observations are central. First, material upward and downward transitions occurred at closely matched frequencies across 78,475 ordinary transitions (rise:fall ratio 0.986), arguing against simple one-directional drift. Second, meal anchoring produced heterogeneous upward, flat and downward responses rather than a uniformly forced spike pattern, with 66.2% of evaluable immediate meal events and 72.6% of broader paired events showing an upward excursion. Third, the 30-minute shifted sensitivity analysis produced a similar, though modestly lower, positive-event proportion (63.5% versus 66.2%), indicating reasonable robustness while making temporal uncertainty explicit. Fourth, 78.0% of evaluable post-peak trajectories subsequently declined. Fifth, after exclusion of Calibration-sheet data and calibration-flagged records, 20 of 21 participants with clean manual/photo fasting and post-meal reference measurements showed the expected higher median post-meal state. Together, these observations support physiological directional responsiveness while defining the limits of the present evidence.

The findings are therefore compatible with physiological trend tracking. They are not, however, equivalent to validation of blood-glucose concentration. This distinction is essential. Directionality can be tested using

physiological event anchors, whereas analytical accuracy requires contemporaneous comparison against a reference glucose method.

Why rise-and-fall detection is a meaningful validation layer

A useful metabolic trend system must respond in both directions. An algorithm that only rises after user interaction, slowly drifts, or systematically reverts to a fixed mean could appear superficially plausible in selected screenshots. The closely balanced counts of >3-unit rises and <-3-unit falls across 78,475 transitions (23,539 rises versus 23,864 falls; ratio 0.986) provide a stronger systems-level observation: the output is capable of two-sided movement over time rather than behaving as a one-sided estimator.

Meal response is an anchor, not an absolute ground truth

Meal logging offers a biologically interpretable event marker, but it is imperfect ground truth. Users may log late, omit snacks, inaccurately estimate portions, or consume mixed meals with fat, protein, fibre, or alcohol that alter the timing and shape of glycaemic response. Large human studies have demonstrated marked person-to-person variability in postprandial glycaemia after identical foods. Accordingly, the scientifically appropriate expectation is an increased probability of an upward response after an adequately carbohydrate-containing meal, not a fixed spike after every meal.

Timing and algorithmic latency

The 30-minute temporal-shift analysis is an important methodological sensitivity test. The proportion of positive meal events changed modestly from 66.2% in the primary immediate analysis (1,395/2,108) to 63.5% after the 30-minute shift (1,349/2,124). The persistence of a majority upward response under the shifted alignment supports reasonable temporal robustness, while the difference also shows that sparse 30-minute sampling and plausible physiological/algorithmic latency affect event anchoring. Reporting both alignments prevents favourable lag selection and makes temporal uncertainty an explicit property of the analysis.

Orthogonal reference measurements strengthen physiological interpretation

The non-calibration reference analysis provides an independent check on the biological context of the cohort. 20 of 21 participants with clean fasting and post-meal reference observations had a higher median post-meal value, with a median within-participant difference of +26 mg/dL. A stricter same-day analysis also showed rises in 39/48 fasting-to-post-meal sequences. Because these measurements were not used as contemporaneous wearable-reference pairs, their appropriate role is orthogonal physiological-state confirmation rather than numerical accuracy validation.

Relationship to prior work

Zeevi and colleagues reported wide variability in glycaemic responses across 46,898 meals in an 800-person cohort, motivating individualized rather than food-only prediction. The PREDICT 1 study similarly demonstrated substantial inter-individual variability in postprandial glucose and showed that person-specific and meal-related factors jointly determine metabolic responses. International CGM consensus work has further established the importance of temporal glucose profiles and standardized longitudinal metrics. The present analysis extends this general concept to a wrist-wearable trend algorithm by treating direction and temporal concordance as an intermediate validation target.

Conceptual innovation: physiological-anchor concordance

The central methodological innovation of this study is a physiological-anchor concordance framework for evaluating wearable-derived metabolic trend algorithms without misrepresenting an inferred signal as a direct biochemical glucose measurement. The framework tests whether the signal satisfies falsifiable, biologically ordered constraints around independently time-stamped real-world events: upward organization after meals, two-sided rise/fall capability during ordinary monitoring, robustness to timing assumptions, and downward recovery after meal-associated peaks.

This framework introduces three complementary properties. Event polarity tests whether the algorithm moves in the physiologically expected direction around an anchor. Bidirectional symmetry tests whether the algorithm is capable of generating rises and falls at comparable population frequency, guarding against monotonic drift or a built-in upward bias. Temporal robustness tests whether conclusions persist under pre-specified alignment shifts, making algorithmic latency visible rather than absorbing it into post-hoc event selection.

These properties define a distinct validation layer for wearable-derived metabolic dynamics. They do not measure numerical blood-glucose accuracy; instead, they test whether a non-glucose wearable signal behaves in a temporally coherent and physiologically interpretable manner. The framework is deliberately falsifiable: a signal dominated by one-directional drift, lacking recovery, or appearing concordant only under one selectively chosen lag would fail these tests.

In the present dataset, the near-unity rise:fall ratio (23,539:23,864; ratio 0.986) provides strong evidence of balanced two-sided responsiveness across 78,475 ordinary transitions. Meal anchoring demonstrates frequent event-associated upward movement, post-peak analysis shows subsequent recovery in 78.0% of evaluable trajectories, and the non-calibration reference analysis adds an orthogonal layer demonstrating expected fasting-to-postprandial state ordering in 20/21 participants. The convergence of these complementary tests is the principal methodological contribution; none is represented as a substitute for contemporaneous reference-glucose accuracy validation.

Strengths

- Real-world free-living data rather than a laboratory-only dataset.
- Repeated measurements across 128 users and 83,090 wearable observations.
- Availability of 3,190 meal logs with time stamps and estimated nutrient composition.
- An orthogonal non-calibration reference-sugar analysis, with the Calibration sheet and calibration-flagged records excluded, demonstrated expected fasting-to-postprandial state ordering independently of the wearable-derived signal.
- Explicit assessment of both rises and falls, reducing the risk of selective “spike-only” interpretation.
- Temporal sensitivity analysis that exposes rather than conceals dependence on lag assumptions.
- Anonymised reporting with no participant names or phone numbers reproduced.

Limitations

- No contemporaneous reference-glucose series was paired to the complete wearable dataset. Therefore, sensitivity, specificity, MARD, Clarke/Parkes error-grid metrics, and numerical glucose accuracy cannot be inferred from this study.
- The orthogonal reference measurements were user-entered or photo-captured and were not used as contemporaneous paired validation of the Rhythm Band-derived output. They support physiological-state interpretation only and should not be interpreted as direct device agreement.
- Meal logs were user-entered and may be delayed, incomplete, or inaccurate. Nutrient estimates were generated at logging time and were not weighed laboratory measurements.
- The data are sparse at approximately 30-minute intervals, so short-lived rises or falls may be missed and exact peak timing cannot be established.
- Multiple observations are nested within users; the exploratory summaries do not model within-person correlation.
- Physical activity, sleep, stress, caffeine, alcohol, medication, illness, and unlogged food intake may confound individual events.
- The exact algorithm version and watch firmware/hardware version should be documented in the final submission for reproducibility.
- The exploratory association between meal composition and excursion is weak; this dataset does not establish proportionality of spike magnitude to carbohydrate or glycaemic load.

- Temporal alignment remains a methodological consideration. The prespecified 30-minute shift reduced the positive meal-event proportion from 66.2% to 63.5%, but a majority of evaluable events remained positive. Exact physiological and algorithmic latency cannot be established from sparse retrospective sampling.

Clinical and Regulatory Interpretation

The present retrospective analysis supports balanced physiological directional responsiveness, frequent meal-associated upward excursions, post-peak recovery, and orthogonal confirmation that the study population exhibited expected fasting-to-postprandial glucose-state ordering. These findings are appropriately interpreted as feasibility evidence for directional metabolic trend inference. They do not establish numerical blood-glucose accuracy, diagnostic performance, or equivalence to CGM or capillary glucose.

Conclusion

In de-identified real-world data from 128 EasyTouch Rhythm Band users operating with the Agatsa One app, the proprietary sugar-trend algorithm produced a strongly bidirectional signal across 78,475 ordinary transitions, with closely matched frequencies of material rises and falls. Meal-linked analyses showed frequent upward excursions and 78.0% of evaluable post-peak trajectories subsequently declined. Independently, after excluding all calibration data, 20/21 participants with clean manual/photo fasting and post-meal reference measurements showed the expected higher post-meal state. These convergent findings support physiological-anchor concordance as a feasibility framework for wearable-derived metabolic trend inference. They do not establish numerical blood-glucose accuracy.

Declarations

Ethics approval and consent to participate. This manuscript reports a retrospective secondary analysis of de-identified real-world records generated during ordinary commercial use of the Agatsa One application and related products. No experimental intervention, study-specific recruitment, participant payment, or study-mandated procedure was performed for this analysis. Before journal submission, the authors should document the applicable institutional ethics/IRB determination (approval, waiver, or exemption, as applicable under the relevant jurisdiction) and insert the committee name and reference number if one was issued.

Consent for publication. No directly identifying participant information, photographs, case narratives, names, telephone numbers, or other identifying details are included in the manuscript. Individual consent for publication of identifiable material is therefore not applicable to the manuscript as written.

Data availability. The analyses are based on de-identified event-level EasyTouch Rhythm Band observations, time-stamped meal logs, and a limited non-calibration reference-sugar stream. The Calibration sheet and calibration-flagged reference records were excluded from the orthogonal reference analysis. A de-identified analytical dataset may be made available subject to privacy, commercial, contractual, and applicable regulatory constraints.

Code and algorithm availability. The Agatsa One sugar-trend algorithm is proprietary software. The analysis definitions required to reproduce the statistical summaries in this manuscript are described in the Methods and Supplementary Appendix. The executable production algorithm and source code are not publicly released because they contain commercially confidential intellectual property. For peer review, sufficient technical documentation, version-locked outputs and analysis code should be made available to editors and reviewers under appropriate confidentiality arrangements where required by journal policy.

Funding. No external or internal study funding was provided for this retrospective analysis. Agatsa Software Pvt. Ltd. did not pay participants, provide study stipends, subsidize participation, or purchase devices for research use. The underlying records arose from ordinary commercial use by customers who purchased Agatsa One subscriptions and/or EasyTouch Rhythm Band products with their own funds. Agatsa developed and operates the commercial platform and proprietary algorithm but did not fund participant participation in this study.

Competing interests. Agatsa One, EasyTouch Rhythm Band, and the sugar-trend algorithm evaluated in this manuscript are commercial/proprietary technologies of Agatsa Software Pvt. Ltd. Authors who are founders, directors, employees, shareholders, consultants, or inventors associated with Agatsa must disclose those relationships explicitly in the final submission. These commercial interests do not constitute study funding; they are reported so readers can assess potential conflicts of interest.

Intellectual property. The manuscript describes a proprietary wearable-derived sugar-trend algorithm and a physiological-anchor concordance framework for evaluating directional metabolic responsiveness. Publication of this manuscript does not by itself determine inventorship, patentability or ownership. Any patent applications, issued patents or pending intellectual-property claims that are materially related to the algorithm or validation framework should be disclosed to the journal as competing interests where applicable and coordinated with publication timing.

Author contributions. Authorship and CRediT roles must reflect actual contributions. The final submission should identify, by author initials, contributions to Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing - original draft; Writing - review & editing; Visualization; Supervision; and Project administration, as applicable. No Funding acquisition role should be assigned for this study because no study funding was provided by Agatsa Software Pvt. Ltd.; any unrelated external support should be disclosed only if it actually contributed to this work.

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Algorithm and device versioning. The manuscript concerns the Agatsa One sugar-trend algorithm operating with EasyTouch Rhythm Band-derived data. Exact Agatsa One application build, production algorithm version/hash, Rhythm Band hardware/firmware version(s), server-side processing version and analysis-script version should be frozen at database lock and reported in the final submission. These identifiers were not contained in the supplied analytical workbook and are therefore not fabricated in this draft.

Reporting transparency. All analyses are retrospective analyses of real-world product-use data. The 30-minute timing sensitivity analysis is reported explicitly. The orthogonal reference analysis excludes the Calibration sheet and calibration-flagged observations and is not presented as numerical device validation.

References

1. Zeevi D, Korem T, Zmora N, et al. Personalized Nutrition by Prediction of Glycemic Responses. *Cell*. 2015;163(5):1079-1094. doi:10.1016/j.cell.2015.11.001.
2. Berry SE, Valdes AM, Drew DA, et al. Human postprandial responses to food and potential for precision nutrition. *Nature Medicine*. 2020;26:964-973. doi:10.1038/s41591-020-0934-0.
3. Battelino T, Danne T, Bergenstal RM, et al. Clinical Targets for Continuous Glucose Monitoring Data Interpretation: Recommendations From the International Consensus on Time in Range. *Diabetes Care*. 2019;42(8):1593-1603. doi:10.2337/dci19-0028.

Supplementary Appendix: Analysis Definitions Used in This Draft

This appendix is included so that the analytical claims in the manuscript are auditable and can be replaced by a locked statistical analysis plan before external submission.

Stored trend value: Midpoint of the source workbook sugar low-high interval.

Raw timestamp primary analysis: Meal time treated as $t=0$ because the workbook specifies that no lag has been applied to stored timestamps.

Timing tolerance: ± 30 minutes where required because median watch sampling interval is 30 minutes.

Post-meal evaluability: At least one pre-meal value in the baseline window and at least one value in the specified post-meal response window.

Material consecutive transition: Absolute change >3 units between consecutive observations separated by 15-90 minutes.

Sensitivity analysis: A 30-minute temporal shift was applied to quantify dependence on potential physiological/algorithmic latency.

No clinical-accuracy endpoint: No MARD, error-grid, or reference-glucose sensitivity/specificity is reported because the present analysis was not designed as a synchronized biochemical accuracy-validation study. The separate non-calibration reference analysis is used only to confirm physiological-state ordering.

Orthogonal reference analysis: The workbook Calibration sheet and all records flagged as calibration references were excluded. The clean reference subset used manual/photo-captured non-calibration observations. Participant-level medians compared fasting with post-meal states; a stricter same-day fasting-to-later-post-meal sequence analysis was supportive.